

$$3 \quad (a) \quad p = \frac{h}{\lambda} = \frac{(6.63 \times 10^{-34})}{(2.4 \times 10^{-10})} = 2.8 \times 10^{-24} \text{ N s}$$

$$(b) \quad E_k = \frac{p^2}{2m_e} = \frac{(2.763 \times 10^{-24})^2}{(2 \times 9.11 \times 10^{-31})} = 4.19 \times 10^{-18} \text{ J}$$

The gain in kinetic energy equals the loss in electric potential energy:

$$E_k = eV = 4.19 \times 10^{-18}$$

$$V = \frac{4.19 \times 10^{-18}}{1.60 \times 10^{-19}} = 26 \text{ V}$$

**Photon energy should not be equated to kinetic energy of the electron.*